

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Viva Voce Answers

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4-AT3 – Viva Voce Answers	Assessment:	CO4-AT3 – Viva Voce Answers
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:	2024	Date:	22-08-2026

Code of Conduct

I A. Reddy Prasad (192472045) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A. Reddy Prasad

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding ; problem not identified correctly	

Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

A. Introduction to Statistical Inference

1. What is statistical inference?

Statistical inference is the process of using **sample data to draw conclusions about a population**.

2. Why is statistical inference important in data science?

It helps us **make decisions and predictions about a population using limited data**.

3. Difference between descriptive and inferential statistics?

- **Descriptive:** Summarizes the given data.
- **Inferential:** Uses sample data to make conclusions about a larger population.

4. What is a population?

A population is the **complete group of individuals or observations** that we are interested in studying.

5. What is a sample?

A sample is a **smaller part of the population** selected for analysis.

6. Why do data scientists work with samples?

Because studying the entire population is often **expensive, time-consuming, or impossible**.

7. What is a population parameter?

A parameter is a **numerical value that describes a population**, such as population mean (μ).

8. What is a sample statistic?

A statistic is a **numerical value calculated from a sample**, such as sample mean ($\bar{x}$).

9. Give a real-world example of statistical inference.

A company surveys **1,000 customers** and uses their responses to estimate the satisfaction of all customers.

10. What are the two major activities in statistical inference?

1. **Estimation**
2. **Hypothesis testing**

B. Frequentist Approach

11. What is the frequentist approach?

It is a statistical approach where probability is based on the **long-run frequency of events**.

12. What is the basic idea behind frequentist statistics?

Use **sample data and repeated-sampling ideas** to make conclusions about unknown population parameters.

13. Frequentist vs Bayesian approach?

- **Frequentist:** Parameter is fixed; data is random.
- **Bayesian:** Parameter is treated as uncertain and can be represented using a probability distribution.

14. What is considered fixed in the frequentist approach?

The **population parameter is fixed**, while the observed data is considered random.

15. What does long-run frequency mean?

It means the probability of an event is the **proportion of times it occurs over many repeated trials**.

16. Why is random sampling important?

It helps ensure that the sample is **representative of the population** and supports valid statistical inference.

17. Can a population parameter be a random variable in frequentist statistics?

No. In the frequentist approach, the population parameter is considered **fixed but unknown**.

18. Give an example of a frequentist problem.

A data scientist uses a sample of customers to test whether a **new recommendation system increases user engagement**.

C. Point Estimation

19. What is a point estimate?

A point estimate is a **single value used to estimate an unknown population parameter**.

Example: Sample mean $\bar{x}$ estimates population mean μ .

20. What is an estimator?

An estimator is a **statistical rule or formula used to estimate a population parameter**.

21. Estimator vs estimate?

- **Estimator:** Formula or method.
- **Estimate:** Numerical result obtained from that formula.

Example: $\bar{x}$ is the estimator; 75 is the estimate.

22. What is the point estimator for population mean?

The **sample mean ($\bar{x}$)**.

23. How do you estimate population proportion?

Using the **sample proportion**:

where x = number of successes and n = sample size.

24. What is the point estimate of population variance?

The **sample variance**:

25. Why is sample mean commonly used to estimate population mean?

Because it is **simple, unbiased, and generally has low variability.**

26. What properties should a good estimator have?

A good estimator should ideally be:

- **Unbiased**
- **Consistent**
- **Efficient**

27. What is an unbiased estimator?

An estimator is unbiased if its **expected value equals the true population parameter.**

28. What is bias?

Bias is the **difference between the expected value of an estimator and the true parameter.**

29. Difference between bias and variance?

- **Bias:** Systematic difference from the true value.
- **Variance:** How much the estimate changes from sample to sample.

30. Why can two samples produce different point estimates?

Because of **sampling variability.** Different random samples contain different observations.

D. Variability in Estimates

31. Why does a point estimate have uncertainty?

Because it is calculated from a **sample rather than the entire population.**

32. What is the sampling distribution of an estimator?

It is the **probability distribution of an estimator over many possible samples** from the same population.

33. What is standard error?

Standard error measures the **variability of an estimator across different samples.**

For the sample mean:

34. Standard error vs standard deviation?

- **Standard deviation:** Measures variability among individual data values.
- **Standard error:** Measures variability of a statistic across samples.

35. How does sample size affect standard error?

As sample size increases, standard error **decreases.**

36. Why does variability decrease when sample size increases?

Larger samples provide **more information about the population**, making estimates more stable.

37. Relationship between standard error and confidence intervals?

A larger standard error produces a **wider confidence interval**, while a smaller standard error produces a narrower interval.

38. If two samples have different means, is one necessarily incorrect?

No. Different random samples can naturally produce different means because of **sampling variability**.

E. Confidence Intervals

39. What is a confidence interval?

A confidence interval is a **range of values used to estimate an unknown population parameter**.

Example:

95% CI = (45, 55)

40. Why is a confidence interval more informative than a point estimate?

A point estimate gives **one value**, while a confidence interval gives both an **estimate and its uncertainty**.

41. What does a 95% confidence level mean?

In the frequentist interpretation, if we repeatedly take samples and construct 95% confidence intervals using the same method, **about 95% of those intervals would contain the true parameter**.

42. Confidence level vs confidence interval?

- **Confidence level:** The reliability level, such as 95%.
- **Confidence interval:** The actual numerical range calculated from the sample.

43. What determines the width of a confidence interval?

Main factors are:

1. Sample size
2. Data variability
3. Confidence level

44. What happens when sample size increases?

The confidence interval generally becomes **narrower**.

45. What happens when confidence level increases from 95% to 99%?

The confidence interval becomes **wider** because we want greater confidence of capturing the true parameter.

46. Difference between z-confidence interval and t-confidence interval?

z-interval

Uses normal distribution

Population SD often known or large-sample setting

Less uncertainty from estimating SD

t-interval

Uses t-distribution

Population SD unknown

Accounts for estimating SD

47. When do you use t-distribution?

Use the **t-distribution** when estimating a population mean and the population standard deviation is **unknown**, especially for smaller samples.

F. Hypothesis Testing Using Confidence Intervals

48. What is hypothesis testing?

Hypothesis testing is a statistical method used to **determine whether there is enough evidence to reject a claim about a population**.

49. What are H_0 and H_1 ?

- **H_0 (Null hypothesis):** The default claim or no-effect assumption.
- **H_1 (Alternative hypothesis):** The claim that we want to find evidence for.

Example:

H_0 : New teaching method has no effect.

H_1 : New teaching method improves performance.

50. How can a confidence interval be used for hypothesis testing?

We compare the **hypothesized parameter value** with the confidence interval.

For a **95% confidence interval**:

- If the hypothesized value is **outside the interval** → **reject H_0** .
- If it is **inside the interval** → **do not reject H_0** .

Quick Viva Revision

Remember these **10 key points**:

Population → Sample → Parameter → Statistic → Estimator → Estimate → Standard Error → Confidence Interval → H_0/H_1 → Decision.